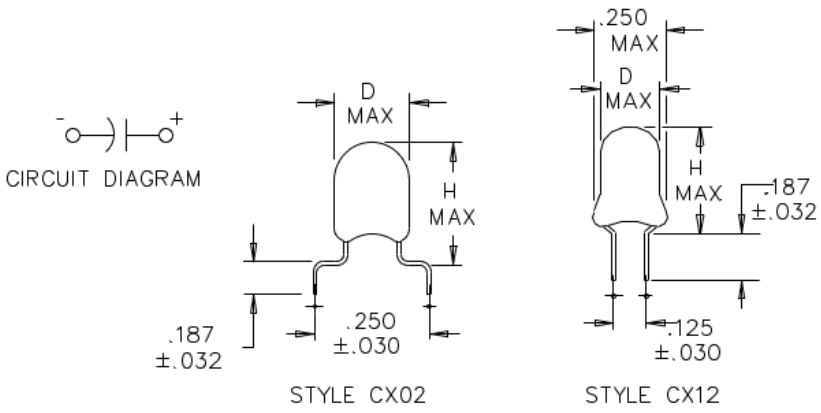
INCH-POUND
MIL-PRF-49137/2F
28 September 2026
SUPERSEDING
MIL-PRF-49137/2E
w/AMENDMENT 1
23 April 2021

PERFORMANCE SPECIFICATION SHEET

# CAPACITOR, FIXED, ELECTROLYTIC (SOLID ELECTROLYTE), TANTALUM, POLAR, CONFORMAL COATED, NONHERMETICALLY SEALED, STYLES CX02 AND CX12

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-49137.




| Dimensions | | | | | |
|---|---|---|---|---|---|
| Style | Case size | Diameter (D) | Height (H) | Lead size | |
| | | | | Dia | AWG |
| CX02 | A | .175 (4.45) | .425 (10.80) | .020 (0.51) | 24, 22 |
| | B | .250 (6.35) | .500 (12.70) | .020 (0.51) | 24, 22 |
| | C | .350 (8.89) | .650 (16.51) | .025 (0.64) | 22 |
| | D | .400 (10.16) | .750 (19.05) | .025 (0.64) | 22 |
| CX12 | E | .175 (4.45) | .350 (8.89) | .020 (0.51) | 24,22 |
| | F | .250 (6.35) | .500 (12.70) | .020 (0.51) | 24,22 |

NOTES:
1. Dimensions are in inches. Metric equivalents are in millimeters and are given for general information only.
2. Metric equivalents are in parentheses.
3. Lead spacing shall be measured within .050 inch (1.27 mm) from the point of emergence from the body or the bottom of the crimp.
4. Case size dimensions are maximum.

FIGURE 1. Dimensions and configuration.



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TABLE I. Style CX02 and CX12 characteristics.

| Part or identifying number (PIN) 1/ | DC rated voltage (+85°C) | Capacitance (nominal) | DC leakage (max) +25°C | Dissipation Factor (DF) | Case size |
|---|---|---|---|---|---|
| | Volts | µF | µA | % | |
| CX - 2D685 - | 6 | 6.8 | 1.0 | 6 | A, E |
| CX - 2D476 - | 6 | 47.0 | 3.0 | 8 | B, F |
| CX - 2D686 - | 6 | 68.0 | 4.5 | 8 | B, F |
| CX02D157 - | 6 | 150.0 | 10.0 | 10 | C |
| CX02D337 - | 6 | 330.0 | 20.0 | 10 | D |
| CX - 2F475 - | 10 | 4.7 | 1.0 | 6 | A, E |
| CX - 2F336 - | 10 | 33.0 | 3.5 | 8 | B, F |
| CX02F107 - | 10 | 100.0 | 10.0 | 10 | C |
| CX02F227 - | 10 | 220.0 | 20.0 | 10 | D |
| CX - 2H335 - | 15 | 3.3 | 1.0 | 6 | A, E |
| CX - 2H226 - | 15 | 22.0 | 3.5 | 8 | B, F |
| CX02H686 - | 15 | 68.0 | 10.0 | 8 | C |
| CX02H157 - | 15 | 150.0 | 20.0 | 10 | D |
| CX - 2J225 - | 20 | 2.2 | 1.0 | 6 | A, E |
| CX - 2J156 - | 20 | 15.0 | 3.0 | 8 | B, F |
| CX02J476 - | 20 | 47.0 | 10.0 | 8 | C |
| CX02J107 - | 20 | 100.0 | 20.0 | 10 | D |
| CX - 2K155 - | 25 | 1.5 | 1.0 | 6 | A, E |
| CX - 2K106 - | 25 | 10.0 | 2.5 | 8 | B, F |
| CX02K336 - | 25 | 33.0 | 9.0 | 8 | C |
| CX02K686 - | 25 | 68.0 | 15.0 | 8 | D |
| CX - 2M685 - | 35 | 6.8 | 2.5 | 6 | B, F |
| CX02M226 - | 35 | 22.0 | 8.0 | 8 | C |
| CX02M336 - | 35 | 33.0 | 10.0 | 8 | D |
| CX02M476 - | 35 | 47.0 | 10.0 | 8 | D |
| CX - 2N104 - | 50 | .1 | 1.0 | 4 | A, E |
| CX - 2N154 - | 50 | .15 | 1.0 | 4 | A, E |
| CX - 2N224 - | 50 | .22 | 1.0 | 4 | A, E |
| CX - 2N334 - | 50 | .33 | 1.0 | 4 | A, E |
| CX - 2N474 - | 50 | .47 | 1.0 | 4 | A, E |
| CX - 2N684 - | 50 | .68 | 1.0 | 4 | A, E |
| CX - 2N105 - | 50 | 1.0 | 1.0 | 4 | A, E |
| CX - 2N155 - | 50 | 1.5 | 1.0 | 6 | B, F |
| CX - 2N225 - | 50 | 2.2 | 1.5 | 6 | B, F |
| CX - 2N335 - | 50 | 3.3 | 2.0 | 6 | B, F |
| CX - 2N475 - | 50 | 4.7 | 2.5 | 6 | B, F |
| CX02N685 - | 50 | 6.8 | 3.5 | 6 | C |
| CX02N106 - | 50 | 10.0 | 5.0 | 8 | C |
| CX02N156 - | 50 | 15.0 | 6.0 | 8 | C |
| CX02N226 - | 50 | 22.0 | 10.0 | 8 | D |

1/ Complete PIN will include additional symbols indicating style and capacitance tolerance.



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REQUIREMENTS

Dimensions and configuration: See figure 1.

Case material: Dipped, epoxy or plastic.

Leads: Solder-coated metal.

DC rated voltage: See table I.

Operating temperature: -55°C to +85°C.

DC leakage (DCL): See table I.

Capacitance (cap.): See table I.

Cap. tolerance: ± 10% (K), or ± 20% (M).

Dissipation factor (DF): See table I.

Stability at low and high temperature: In accordance with MIL-PRF-49137.

Step 1 (+25°):

DCL: See table I.
Cap.: Within tolerance specified in table I.
DF: See table I.

Step 2 (-55°C):

ΔCap.: Within ± 15 percent of step 1 measured value.
DF: 200 percent of table I.

Step 3 (+25°C):

DCL: See table I.
ΔCap.: Within ± 10 percent of step 1 measured value.
DF: See table I.

Step 4 (+85°C):

DCL: 10 times +25°C limit (see table I).
ΔCap.: Within ± 15 percent of step 1 measured value.
DF: 150 percent of table I.

Step 5 (+25°C):

DCL: See table I.
ΔCap.: Within ± 10 percent of step 1 measured value.
DF: See table I.

Surge voltage: In accordance with MIL-PRF-49137.

DCL: See table I.
ΔCap.: Within ± 10 percent of initial value.
DF: See table I.



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Life (1,000 hours at +25°C): In accordance with MIL-STD-202-108.

DCL: 200 percent of table I.
ΔCap.: Within ± 10 percent of initial measured value.
DF: See table I.

NOTE: These capacitors are intended to be used only where supplemental moisture protection is provided or for noncritical applications where hermetic moisture protection is not required.

Reference documents. In addition to MIL-PRF-49137, this specification sheet references the following:

MIL-STD-202-108

Changes from previous issue: The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy – EC
Air Force – 85
DLA – CC

Review activities:
Navy - AS, MC
Air Force - 19

Preparing activity:
DLA - CC

(Project 5910-2026-046)

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